Why the "+1" in the shifted cosine builds an item-bias channel

A fully worked numeric demonstration ($B(t)$ on the U-ProtoMF host)

2026-07-12 — dc05 review companion

1. Setup

U-ProtoMF with $K = 3$ user prototypes. A user u has the cosine vector $c(u) = \big(\cos(q_u, p_1), \cos(q_u, p_2), \cos(q_u, p$ and the *shifted* activations are $u^* = \mathbf{1} + c(u)$, i.e. each entry is $1 + \cos$. An item is a free vector $t \in \mathbb{R}^3$. The score is

$$S(u,t) \;=\; t^\top u^* \;=\; t^\top\big(\mathbf{1} + c(u)\big) \;=\; \underbrace{t^\top \mathbf{1}}_{B(t)\ \text{(no } u \text{ anywhere)}} \;+\; \underbrace{t^\top c(u)}_{\text{personalized}} .$$

The split is trivial algebra — the point of this note is what the blue term can *do*.

2. The demonstration

Two users and two items, numbers chosen so you can check every line by hand:

$$c(\text{Alice}) = (0.6,\ -0.3,\ 0.0), \qquad c(\text{Bob}) = (-0.2,\ 0.4,\ 0.1).$$

$t_A = (1.0,\ 0.5,\ 0.8), \qquad t_B = t_A + v \quad \text{with} \quad v = (-0.3,\ -0.6,\ 1.8), \qquad \text{so } t_B = (0.7,\ -0.1,\ 2.6).$

The direction v was constructed with two properties (verify by hand):

$v^\top c(\text{Alice}) = -0.18 + 0.18 + 0 = 0, \qquad v^\top c(\text{Bob}) = 0.06 - 0.24 + 0.18 = 0, \qquad v^\top \mathbf{1} = 0.9 \neq 0.$

So items A and B are *identically attractive* to both users in every taste dimension the users actually differ in — v is invisible to both cosine vectors — and the items differ *only* along the all-ones direction.

Scores with the shift (the real model)

	$B(t) = t^\top \mathbf{1}$	$t^\top c(u)$	$S = B + \text{pers.}$	
Alice, item A	2.3	0.45	2.75	
Alice, item B	3.2	0.45	3.65	$\Leftarrow$ wins by 0.9
Bob, item A	2.3	0.08	2.38	
Bob, item B	3.2	0.08	3.28	$\Leftarrow$ wins by 0.9

Check one cell by hand, e.g. Alice/item B personalized: $0.7 \cdot 0.6 + (-0.1)(-0.3) + 2.6 \cdot 0 = 0.42 + 0.03 + 0 = 0.45$.

Item B beats item A **for every user, by exactly the same margin** $v^\top \mathbf{1} = 0.9$, although no user's taste distinguishes them. That margin is a pure per-item, user-independent score component — *by definition, an item bias* — and the model can set it freely by moving t along the all-ones direction.

Now delete the "+1" (unshifted cosine)

The score becomes $S = t^\top c(u)$ only:

$$\text{Alice: } 0.45 \text{ vs } 0.45 \quad \text{(exact tie)}, \qquad \text{Bob: } 0.08 \text{ vs } 0.08 \quad \text{(exact tie)}.$$

The *same* two items, the *same* direction v — and the channel is gone. Without the shift, a direction that no user's cosine vector sees contributes **nothing** to any score: it is an invisible (gauge) direction. With the shift, that same direction contributes $v^\top \mathbf{1}$ to **everyone's** score: it has become a universal dial.

> **The mechanism in one sentence:** the +1 inserts a constant coordinate into every user's representation ($u^* = \mathbf{1} + c$), and in a linear score a constant input column is an intercept — so the free item vector's component along $\mathbf{1}$ becomes a learned per-item intercept $B(t)$, exactly like the intercept column in linear regression. The shift *converts an invisible direction into a universal one.*

3. Why fI's "+1" is harmless (the mirror)

The shift always creates the intercept; the question is *whose* it is. The constant sits in the *composed* side's activations, so the intercept coefficient belongs to the *free* side's entity:

$$\text{fI: } S = u^\top(\mathbf{1}+c(i)) = \underbrace{u^\top \mathbf{1}}_{\text{per-USER intercept}} + u^\top c(i), \qquad \text{fU: } S = t^\top(\mathbf{1}+c(u)) = \underbrace{t^\top \mathbf{1}}_{\text{per-ITEM intercept}} + t^\top c(u).$$

Ranking runs *across items for a fixed user*. A per-user intercept (fI) adds the same number to every item that user ranks — it can never reorder them (rank-inert). A per-item intercept (fU) differs between the items being ranked — it reorders them (live). Same "+1", opposite consequence, decided purely by which side is free.

4. Two honest footnotes

(a) The exact-$\perp$ construction is for clarity, not realism. With only two users, a direction orthogonal to both cosine vectors trivially exists. In a trained model with many users the *exactly* invisible directions exist whenever $K_u > d + 1$ (the gauge space); but the bias channel itself needs no orthogonality at all — *any* movement of t along $\mathbf{1}$ shifts $B(t)$; the $\perp$ trick above only serves to isolate the channel from taste effects so the demonstration is airtight.

(b) The paper's stated reason for the shift is innocent. The $1 + \cos$ form is introduced to keep activations positive in $[0, 2]$ (ProtoMF §3.1). The intercept is an unintended structural side effect — present in published U-ProtoMF, never separated before. On the I host the same shift is rank-inert, which is why "the fleet is bias-free" was a correct working statement for fI and needs the per-host footnote from here on.